

Future Media Internet

Guoping Tan

Email: gptan@hhu.edu.cn

2023. 10. 20

Main Content

- **Background**
- **Data Communications and Networks**
- **OSI Model & TCP/IP**
- **Physical Layer**
- **MAC Layer**
- **Multimedia Communications**

Use Cases on Holographic-Type Communications



(a) Hologram lectures at Imperial College London [26]. The life-size holographic speakers in the middle are projected onto a display.



(b) Hologram curriculum developed by Pearson and Microsoft is used to train nurses [27].



(c) A doctor uses the holographic display during a cardiac ablation procedure [28].



(d) 3D reconstruction of a soccer game using XR HMDs [29].



(e) Boxing broadcast developed by Condense Reality [30].



(f) Axiom Holographics (formerly Euclidean Holographics) hologram arcade table [31], [32].

Source: Ian F. Akyildiz and Hongzhi Guo, Holographic-Type Communication: A New Challenge for the Next Decade

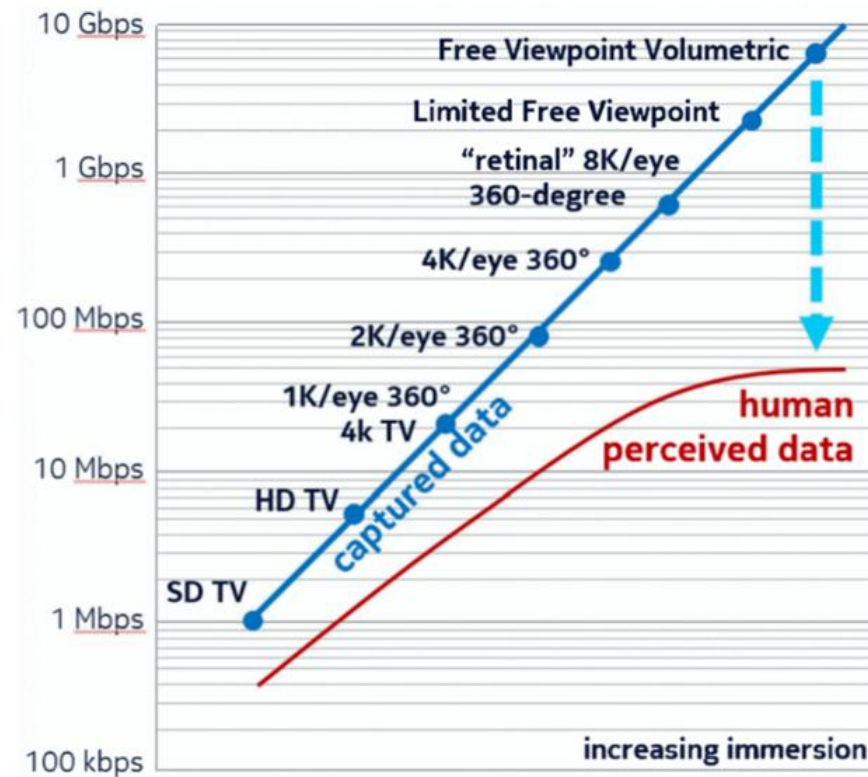
Video: <https://aexa.com/holowizard/overview/>

Quality of service requirements for VR phases

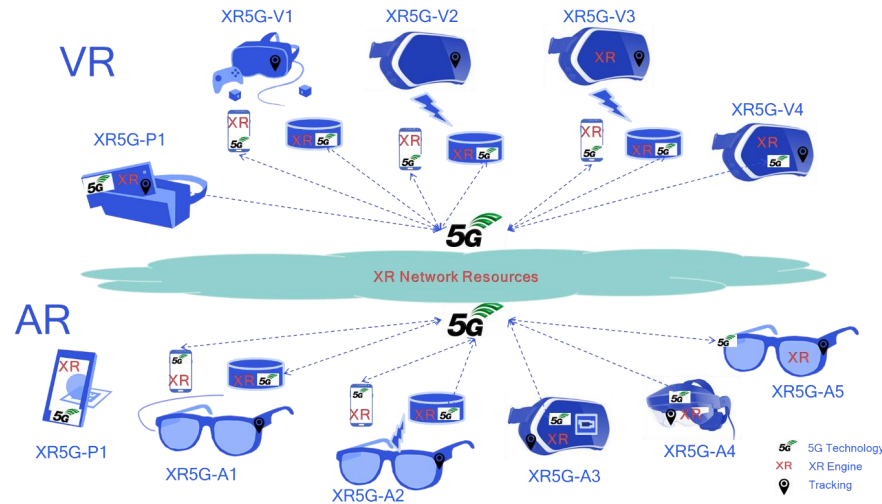
Requirement		Pre-VR	Entry-level VR	Advanced VR	Human perception	Ultimate VR
Experience duration		Less than 20 minutes	Less than 20 minutes	Less than an hour	/	More than an hour
Video resolution		3840 X S1920 (full-view 4K video)	7680 x 3840 (full-view 8K video)	11,520 x 5760 (full-view 12K video)	21,600 x 10,800 (full-view video)	23040 x 11520 (full-view 24K video)
Single-eye resolution		1080 x 1080	1920 x 1920	3840 x 3840	9000 x 8100	9600 x 9600
Field of view (single-eye)		100 x 100	110 x 110	120×120	150×135	150 x 150
Bit per color (RGB)		8	8	10	/	12
Refresh rate		60	90	120	120	200
Pixel per degree		10	17	32	60	64
Service Requirement	Uncompressed bit rate (progressive 1:1)*	10.62 Gb/s	63.70 Gb/s	238.89 Gb/s	1007.77 Gb/s	1911.03 Gb/s
	Transmitting bit rate (low-latency compression 20:1)	530 Mb/s	3.18 Gb/s (Full-view) 796 Mb/s (FoV)	11.94 Gb/s (Full-view) 5.31 Gb/s (FoV)	50.39 Gb/s (Full-view) 31.49 Gb/s (FoV)	95.55 Gb/s (Full-view) 66.36 Gb/s (FoV)
	Transmitting bit rate (lossy compression 300:1)	35 Mb/s	210 Mb/s (Full-view) 53 Mb/s (FoV)	796 Mb/s (Full-view) 354 Mb/s (FoV)	3.36 Gb/s (Full-view) 2.10 Gb/s	6.37 Gb/s (Full-view) 4.42 Gb/s (FoV)
	Typical Round-trip time (RTT)	10 ms	10ms	5ms	10ms	5ms
	Typical packet loss	1e-6	1e-6	1e-6	1e-6	1e-6

* Progressive data rate = (3 x bit per color) x (pixel per degree x field of view (full-view or single eye)) x refresh rate/compression ratio

Human perceived data rate while streaming video VS. captured data rate



XR Form Factors (3GPP TR26.928)



XR Type Number	XR Device Type Name	Tethering Examples	5G Uu Modem	XR Engine Localization	Power Supply	Typical Max Avail Power
XR5G-P1	Phone	n/a	XR device	XR device or split	Internal	3-5 W
XR5G-V1	Simple VR Viewer wired tethering	USB-C	External	External	External	2-5 W
XR5G-V2	Simple VR Viewer wireless tethering	802.11ad/y, 5G sidelink, etc.	External	External	Internal	2-3 W
XR5G-V3	Smart VR Viewer wireless tethering	802.11ad/y, 5G sidelink, etc.	External	XR device or Split	Internal	2-3 W
XR5G-V4	VR HMD Standalone	n/a	XR device	XR device or Split	Internal	3-7 W
XR5G-A1	Simple AR Wearable Glass wired tethering	USB-C	External	External	External	1-3 W
XR5G-A2	Simple AR Wearable Glass wireless tethering	802.11ad/y, 5G sidelink, etc.	External	External	Internal	0.5 – 2 W
XR5G-A3	Smart AR HMD see-through standalone	n/a	XR device	XR device or Split	Internal	3-7 W
XR5G-A4	AR Wearable Glass standalone	n/a	XR device	XR device or Split	Internal	2 - 4 W
XR5G-A5	Smart AR Wearable Glass wireless tethering	802.11ad/y, 5G sidelink, etc.	External	XR device or Split	Internal	0.5 – 2 W

Future HTC Development

Table 1 – Projected future HTC development.

	Current	5 Years	10 Years (6G & beyond)
Data Rates	1 Gbps	100 Gbps	>1 Tbps
Refresh Rates	≤ 30 fps	60 fps	≥ 90 fps
Distance	Local Area	Metropolitan	Very Wide Area

Videos → VR → MR&AR → HTC

Table 2 – Comparison of technical requirements of videos, VR, AR and MR, and HTC.

	Video	VR	MR & AR	HTC
Data Rates	35-140 Mbps	25 Mbps-5 Gbps	25 Mbps-5Gbps	1-10 Tbps
Latency	15-35 ms	<20 ms	<10 ms	<1 ms
Synchronization	Audio/Video	Multiple Tiles	Multiple Tiles	Thousands Views & Senses
Devices	Smart Devices/ Computers	Head-Mounted Displays	Head-Mounted Displays (Smart Glasses)	Naked Eyes or Head-Mounted Displays

Lecture 1

Data Communications and Networks

Outline

- **1-1 DATA COMMUNICATIONS**
- **1-2 NETWORKS**
- **1-3 PROTOCOLS AND STANDARDS**

1-1 DATA COMMUNICATIONS

*The term **telecommunication** means communication at a distance. The word **data** refers to information presented in whatever form is agreed upon by the parties creating and using the data. **Data communications** are the exchange of data between two devices via some form of transmission medium such as a wire cable.*

Topics discussed in this section:

Components

Data Representation:

(Text,Numbers,Images, Audo, Video etc.)

Data Flow

Figure 1.1 *Five components of data communication*

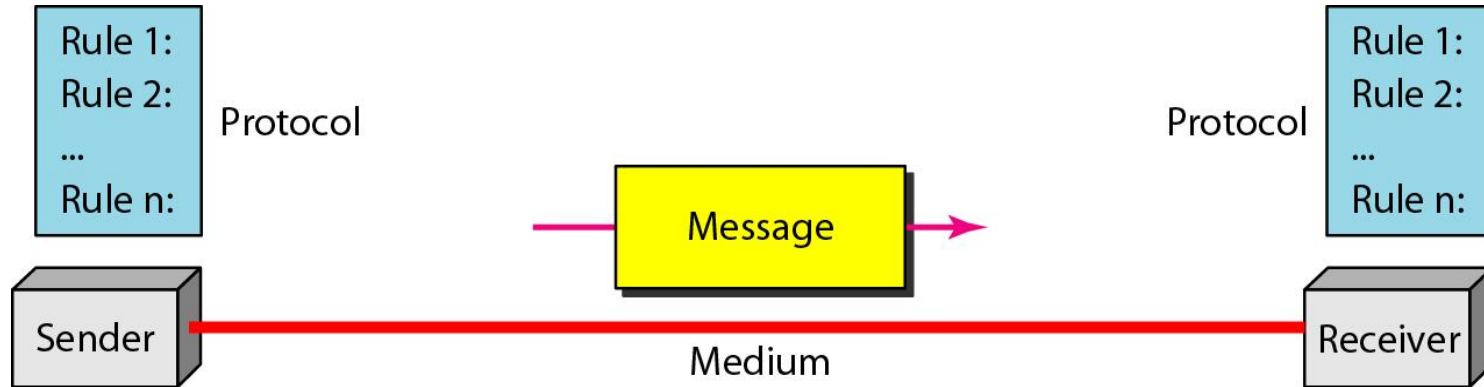
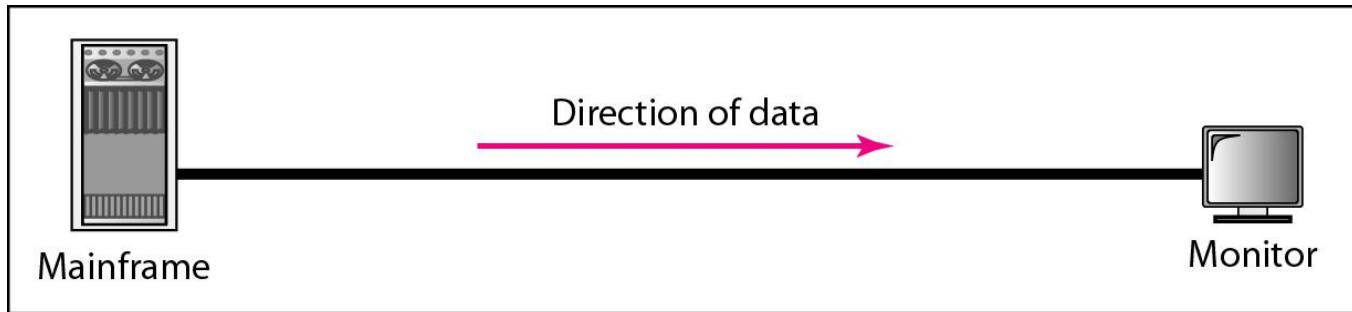
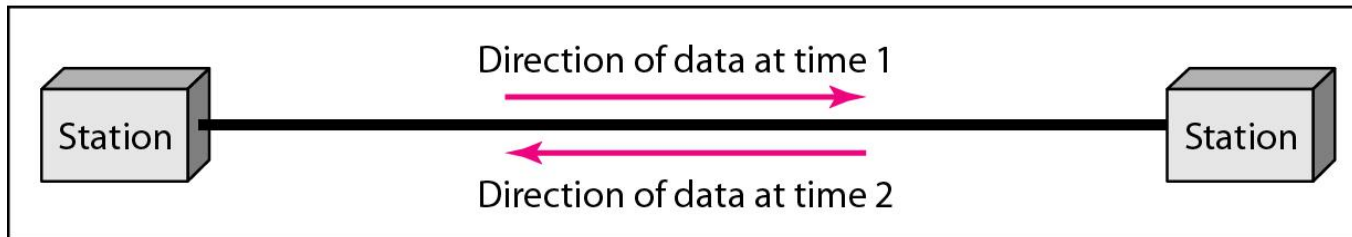


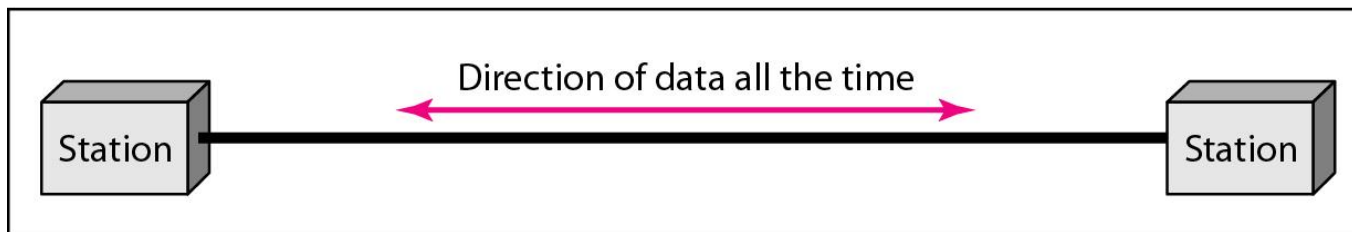
Figure 1.2 *Data flow (simplex, half-duplex, and full-duplex)*



a. Simplex



b. Half-duplex



c. Full-duplex

1-2 NETWORKS

*A **network** is a set of devices (often referred to as **nodes**) connected by communication **links**. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.*

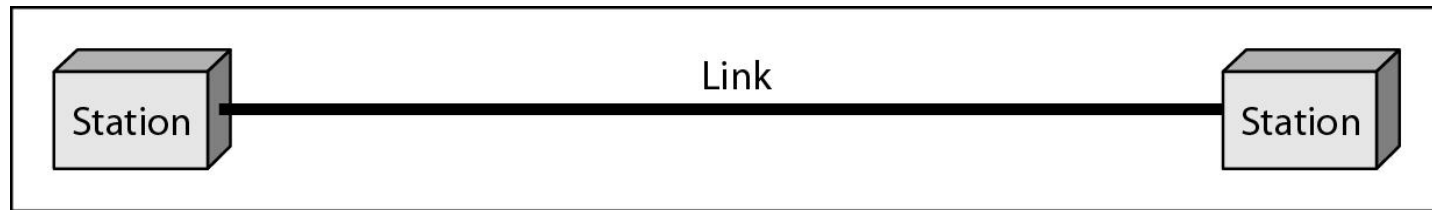
Topics discussed in this section:

Network Models

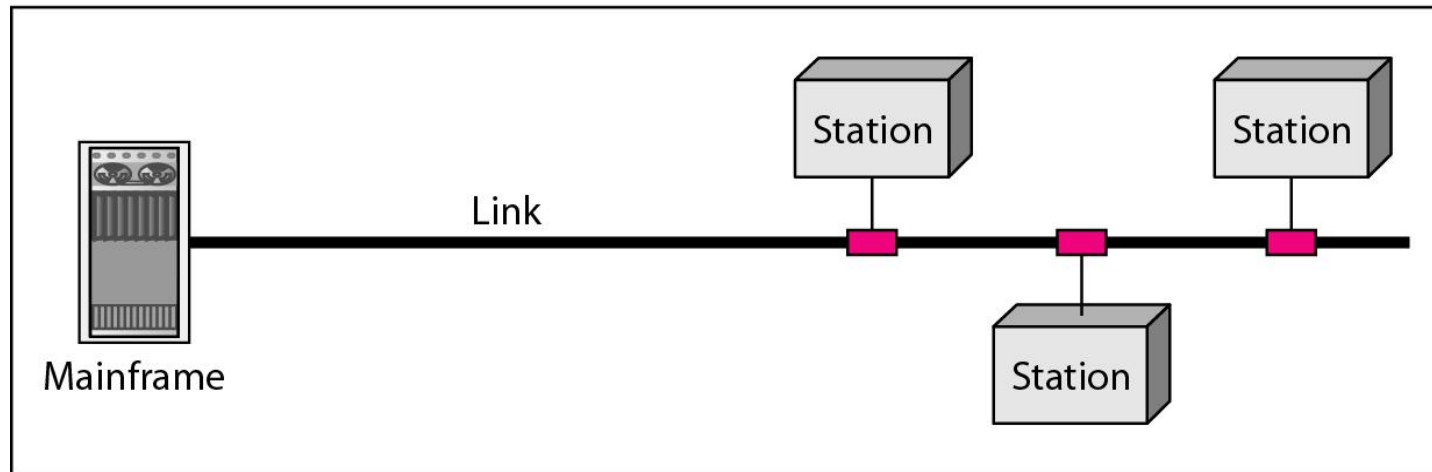
Categories of Networks

Interconnection of Networks: Internetwork

Figure 1.3 *Types of connections: point-to-point and multipoint*



a. Point-to-point



b. Multipoint

Figure 1.4 *Categories of topology*

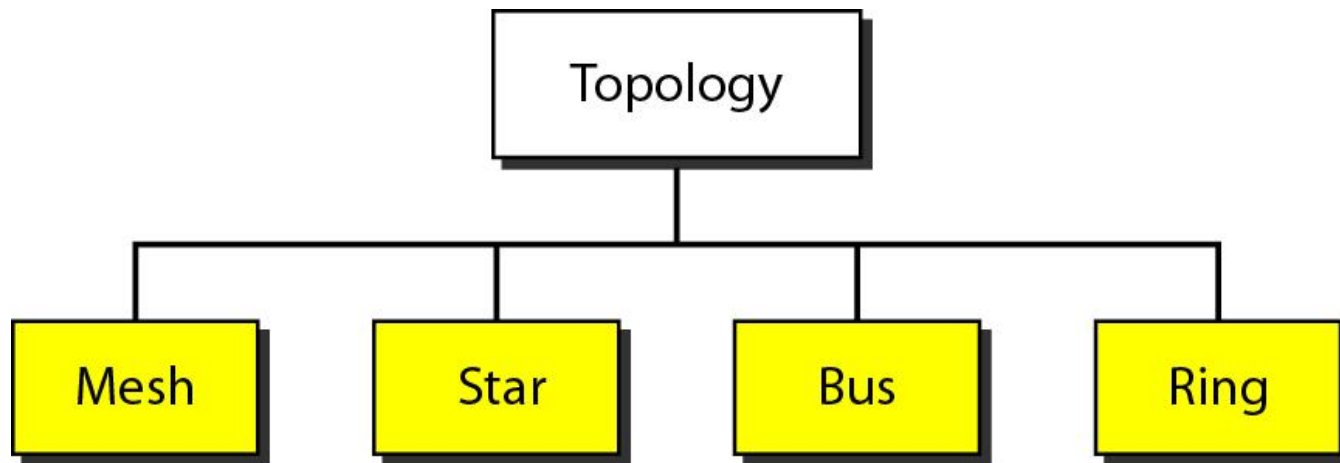


Figure 1.5 *A fully connected mesh topology (five devices)*

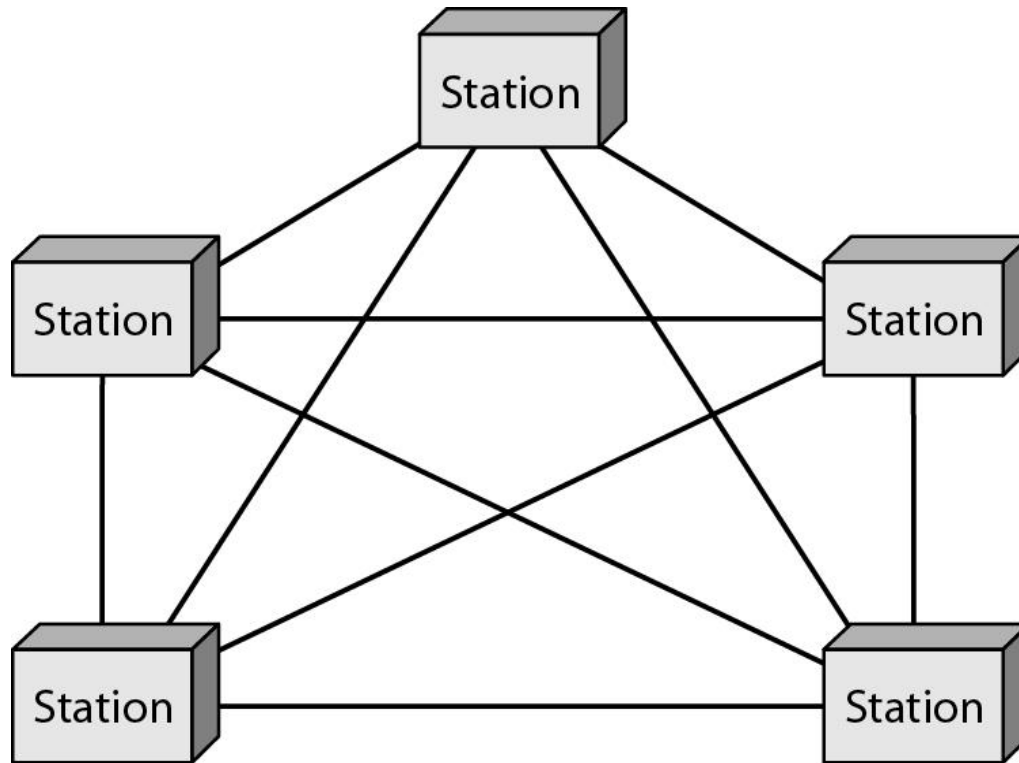


Figure 1.6 *A star topology connecting four stations*

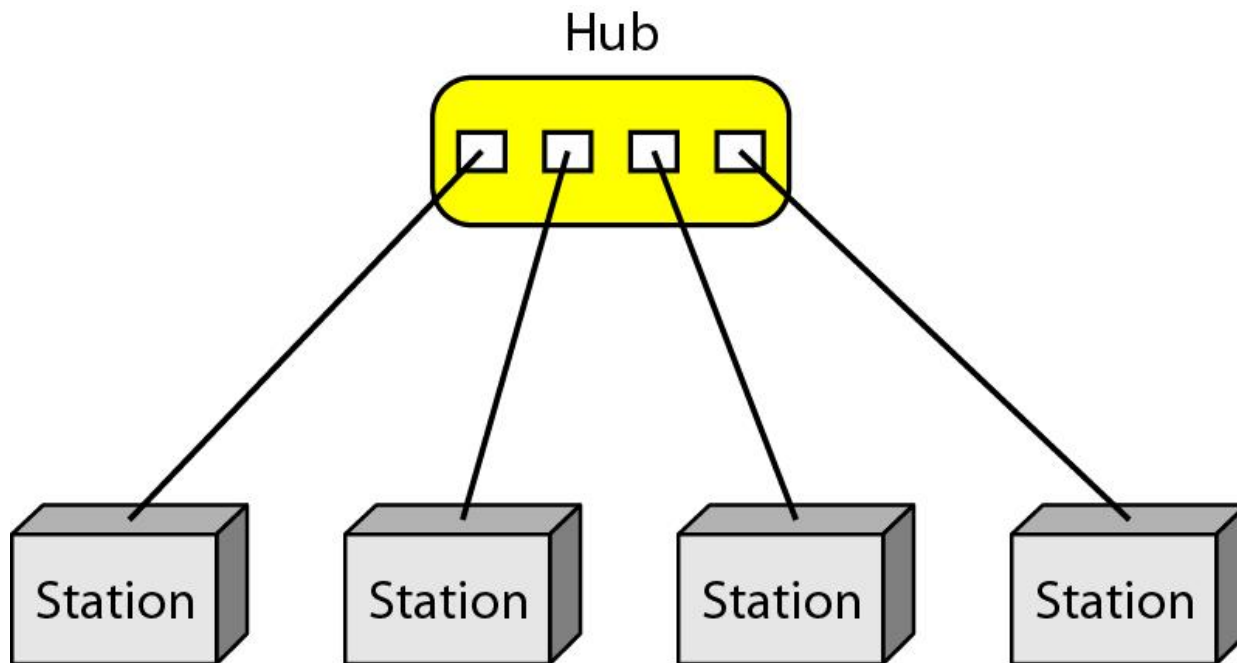


Figure 1.7 *A bus topology connecting three stations*

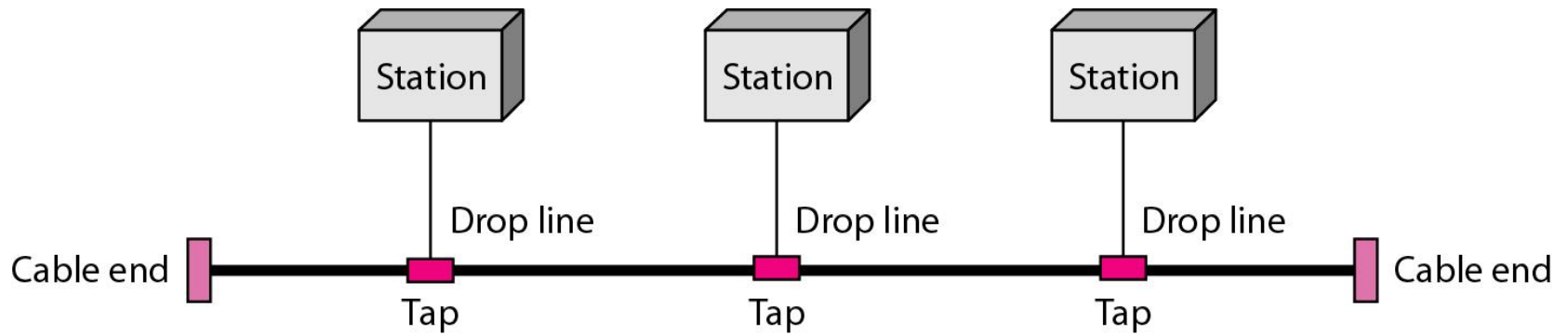


Figure 1.8 *A ring topology connecting six stations*

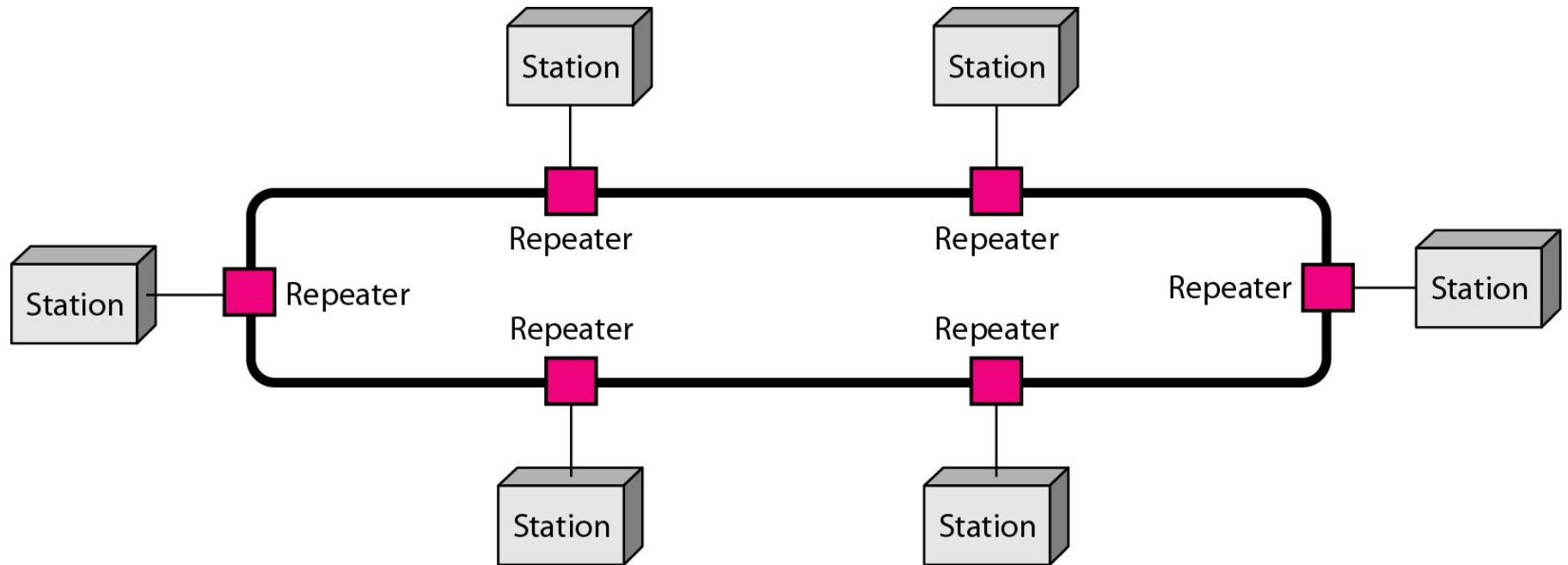
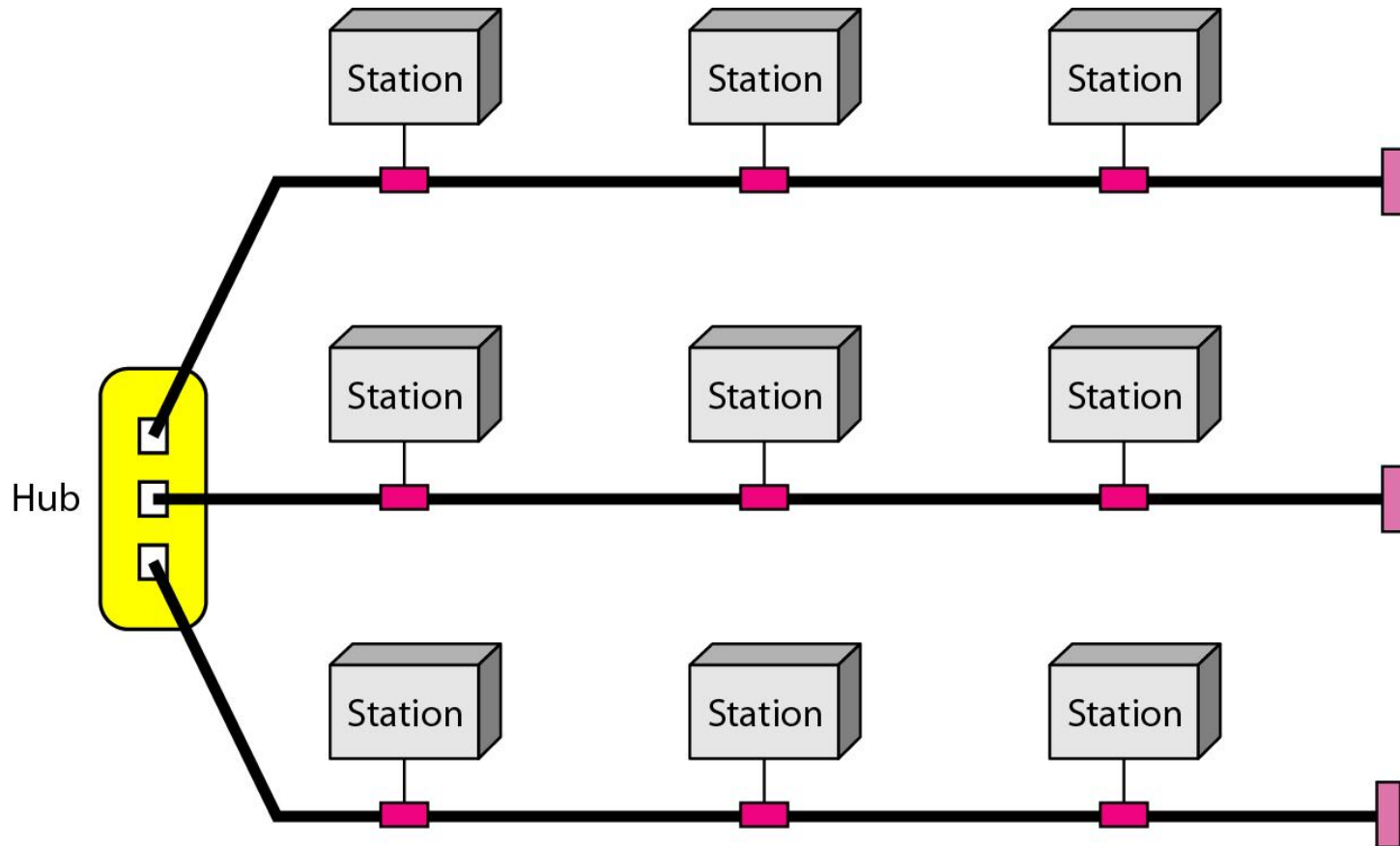
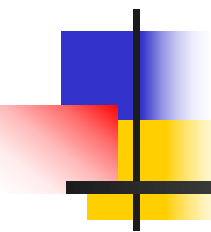


Figure 1.9 *A hybrid topology: a star backbone with three bus networks*

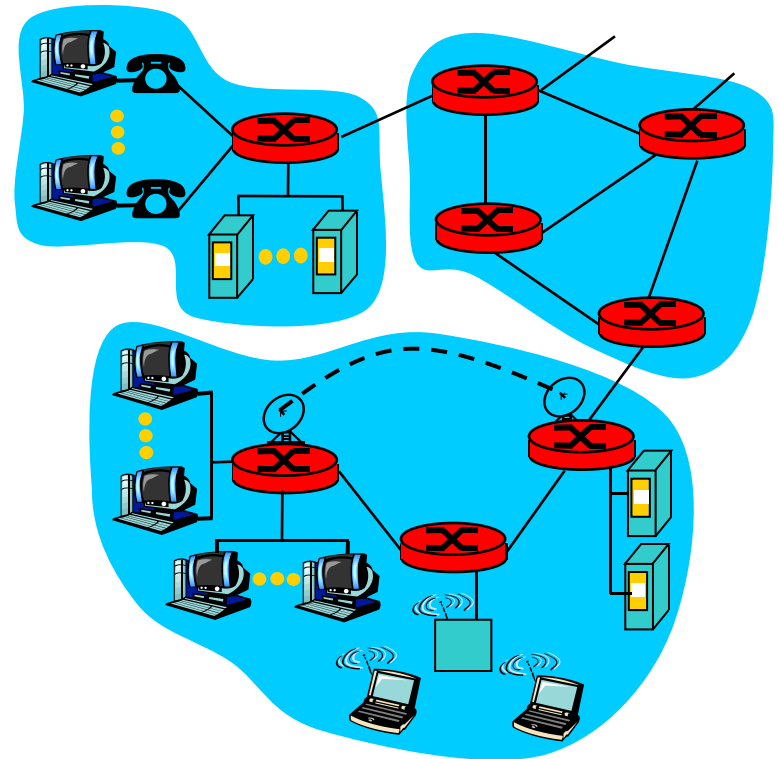




A Closer Look at Network Structure

A closer look at network structure:

- **network edge:**
applications and hosts
- **network core:**
 - routers
 - network of networks
- **access networks, physical media:** communication links



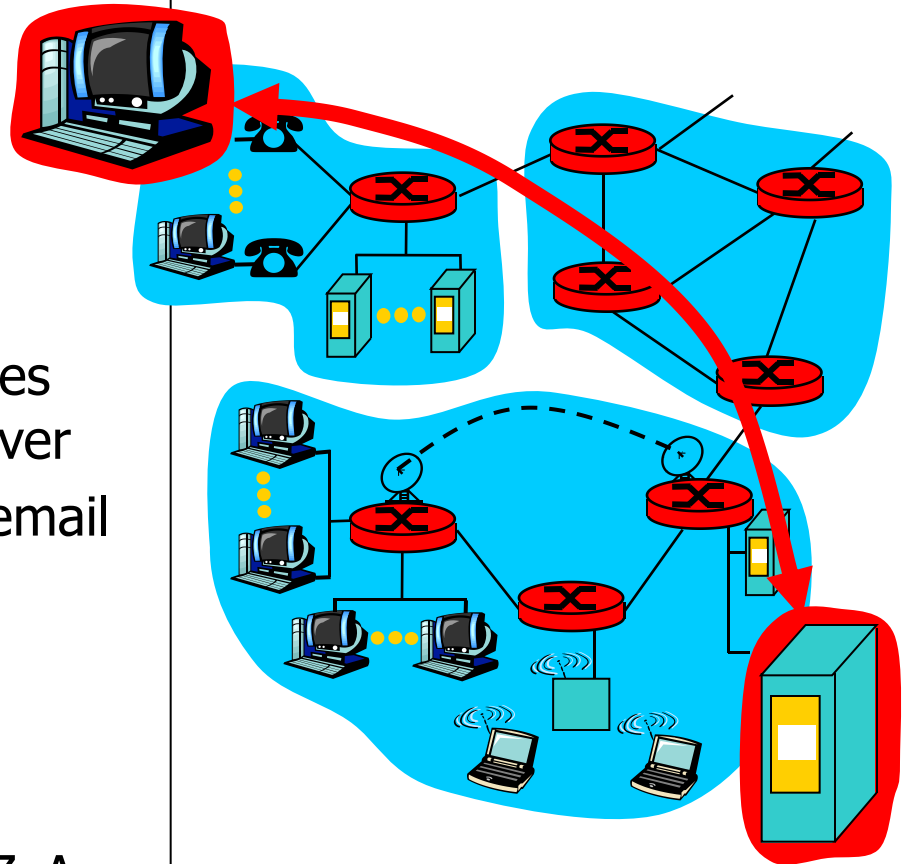


A Closer Look at Network Structure:

Network Edge

The network edge:

- **end systems (hosts):**
 - run application programs
 - e.g. Web, email
 - at “edge of network”
- **client/server model**
 - client host requests, receives service from **always-on** server
 - e.g. Web browser/server; email client/server
- **peer-peer model:**
 - minimal (or no) use of dedicated servers
 - e.g. Skype, BitTorrent, KaZaA



Network edge: connection-oriented service

Goal: data transfer between end systems

- *handshaking:* setup (prepare for) data transfer ahead of time
 - Hello, hello back human protocol
 - *set up "state"* in two communicating hosts
- TCP - Transmission Control Protocol
 - Internet's connection-oriented service

TCP service [RFC 793]

- *reliable, in-order* byte-stream data transfer
 - loss: acknowledgements and retransmissions
- *flow control:*
 - sender won't overwhelm receiver
- *congestion control:*
 - senders "slow down sending rate" when network congested

Network edge: connectionless service

Goal: data transfer between end systems

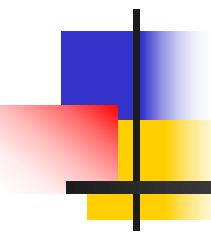
- same as before!
- **UDP** - User Datagram Protocol [RFC 768]:
 - connectionless
 - unreliable data transfer
 - no flow control
 - no congestion control

App's using TCP:

- HTTP (Web), FTP (file transfer), Telnet (remote login), SMTP (email)

App's using UDP:

- streaming media, teleconferencing, DNS, Internet telephony

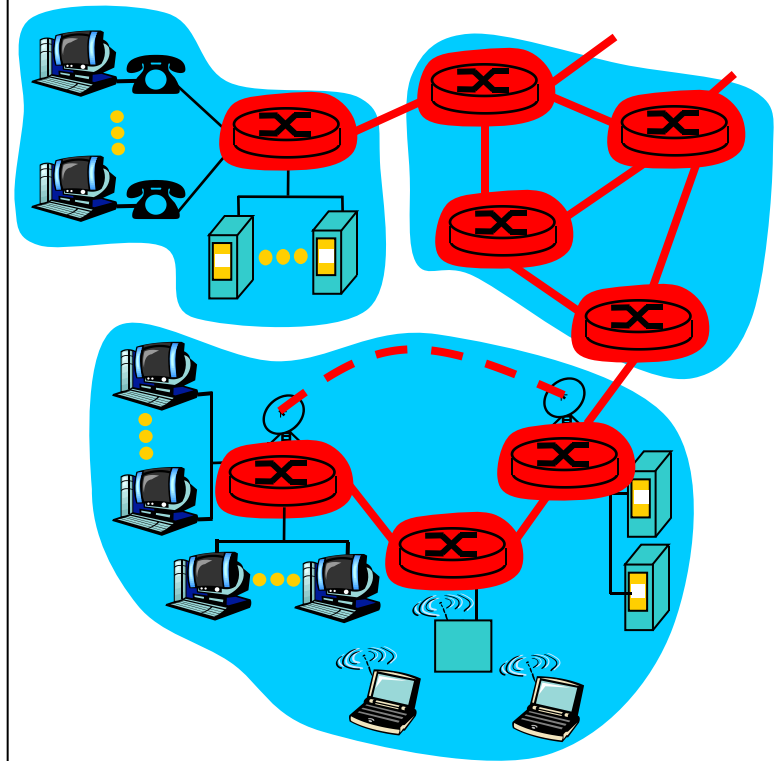


A Closer Look at Network Structure:

Network Core

The Network Core

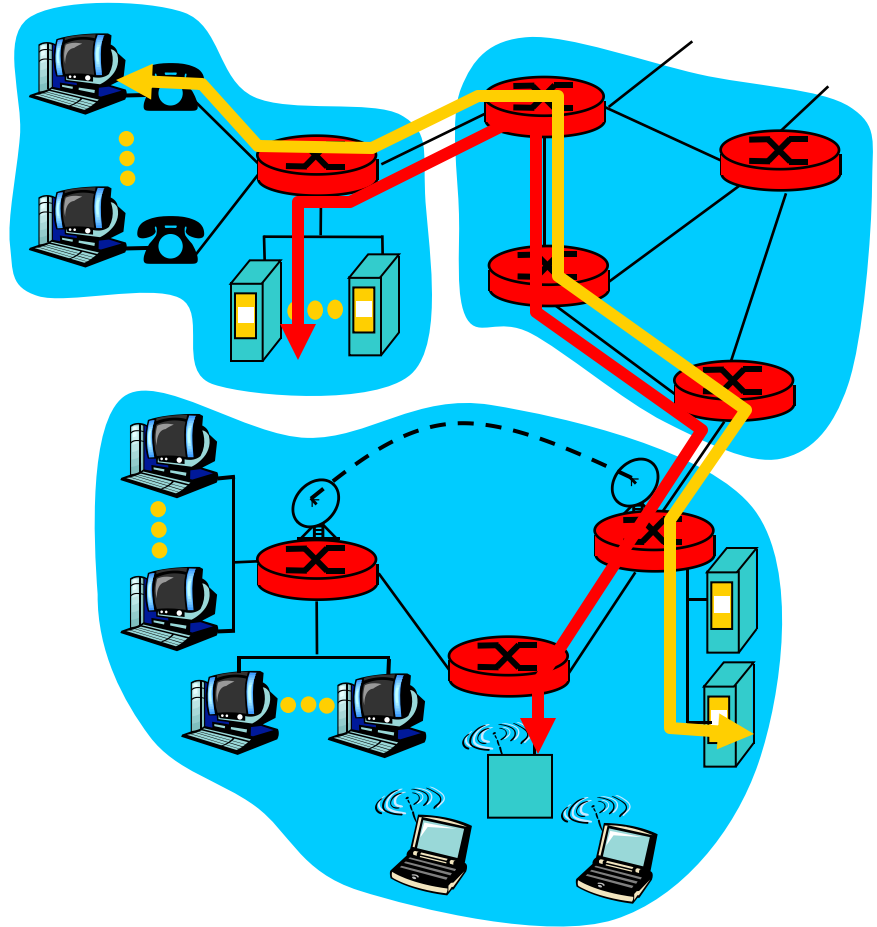
- Mesh of interconnected routers
- *the fundamental question:* how is data transferred through net?
 - **circuit switching:** dedicated circuit per call: telephone net
 - **packet-switching:** data sent thru net in discrete "chunks"



Network Core: Circuit Switching

End-end resources reserved for “call”

- link bandwidth, switch capacity
- dedicated resources: no sharing
- circuit-like (guaranteed) performance
- call setup required



Network Core: Circuit Switching

network resources (e.g., bandwidth) divided into "pieces"

- pieces allocated to calls
- resource piece *idle* if not used by owning call (*no sharing*)

- dividing link bandwidth into "pieces"
 - frequency division (FDM)
 - time division (TDM)
 - SDM

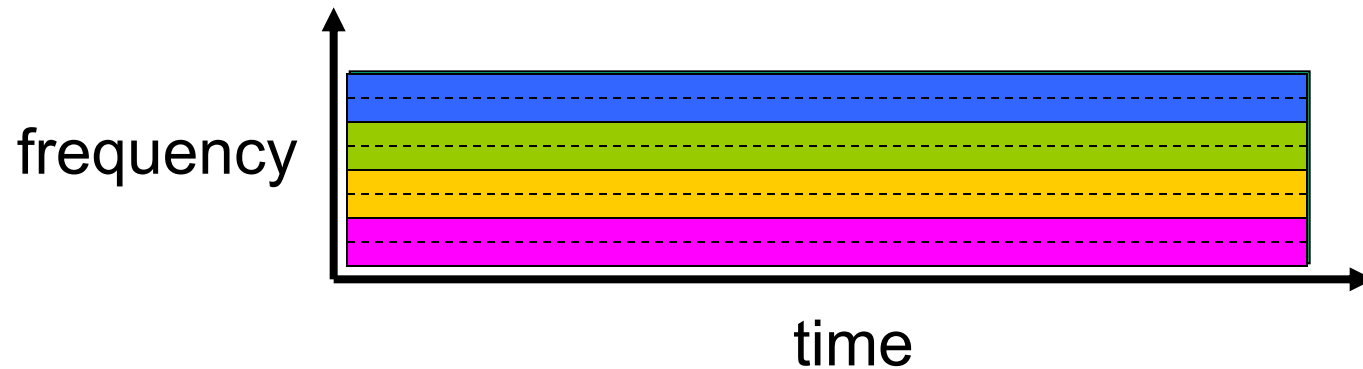
Circuit Switching: FDM and TDM

Example:

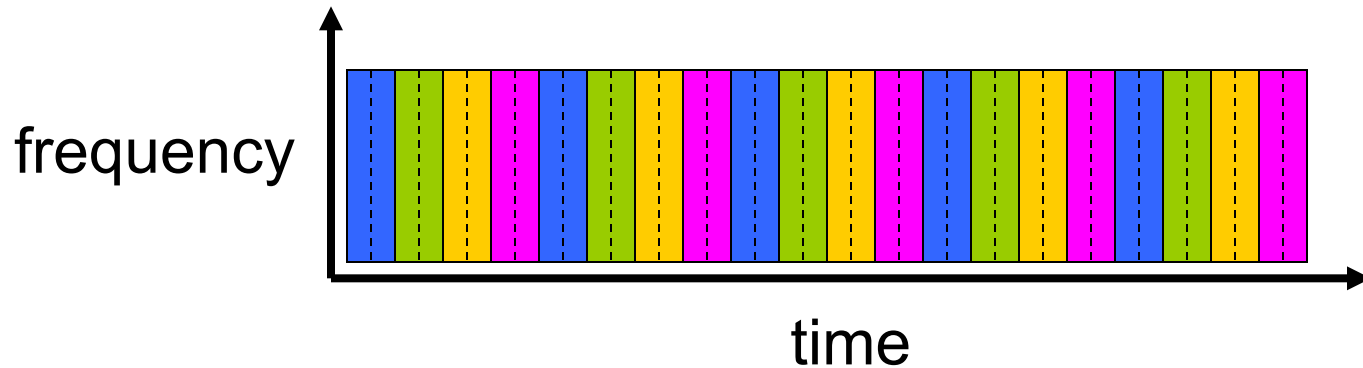
4 users



FDM



TDM



Network Core: Packet Switching

each end-end data stream
divided into *packets*

- user A, B packets *share* network resources
- each packet uses full link bandwidth
- resources used *as needed*

resource contention:

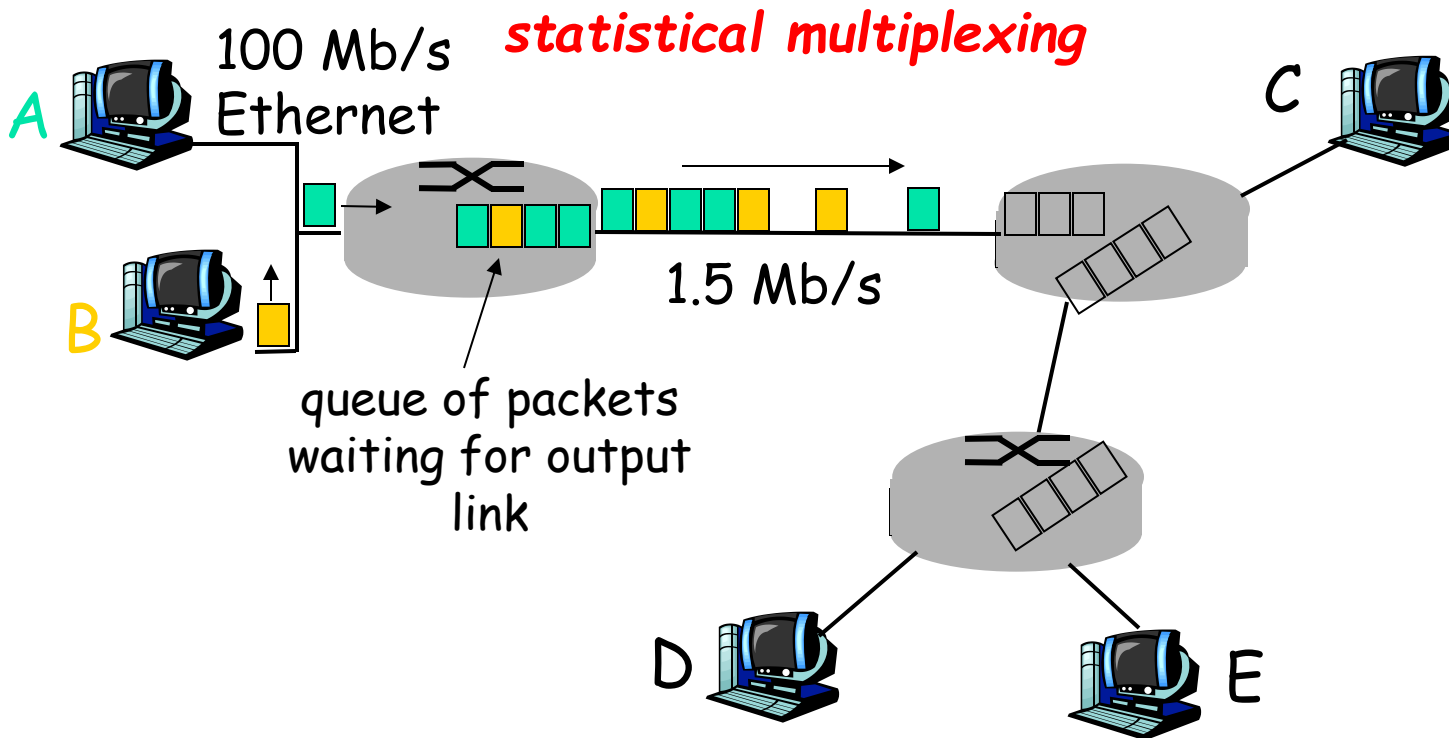
- Aggregate resource demand can exceed amount available
- congestion: packets queue, wait for link use
- store and forward: packets move one hop at a time
 - Node receives complete packet before forwarding

Bandwidth division into
“pieces”

Dedicated allocation

Resource reservation

Packet Switching: Statistical Multiplexing



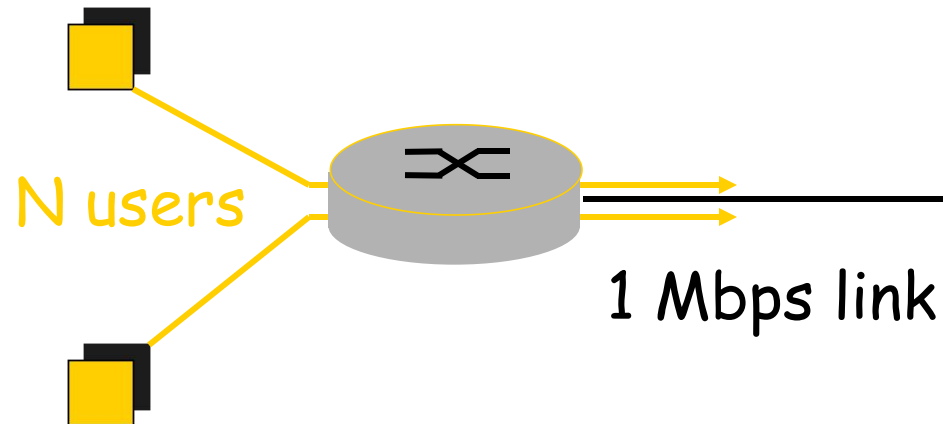
Sequence of A & B packets does not have fixed pattern,
shared on demand → **statistical multiplexing**.

TDM: each host gets same slot in revolving TDM frame.

Packet switching vs. circuit switching

Packet switching allows more users to use network!

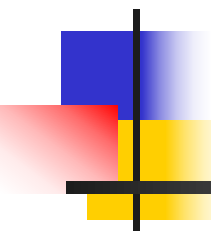
- 1 Mb/s link
- each user:
 - 100 kb/s when “active”
 - active prob. = 0.1 for each time slot
- circuit-switching:
 - 10 users
- packet switching:
 - with 35 users,
Prob. (active users > 10)
less than .0004



Packet switching vs. circuit switching

Is packet switching a “slam dunk winner?”

- Great for bursty data
 - resource sharing
 - simpler, no call setup
- **Excessive congestion:** packet delay and loss
 - needing protocols for reliable data transfer
 - congestion control
- **Q: How to provide circuit-like behavior?**
 - bandwidth guarantees needed for audio/video apps
 - still an unsolved problem



A Closer Look at Network Structure:

Access Networks and Physical Media

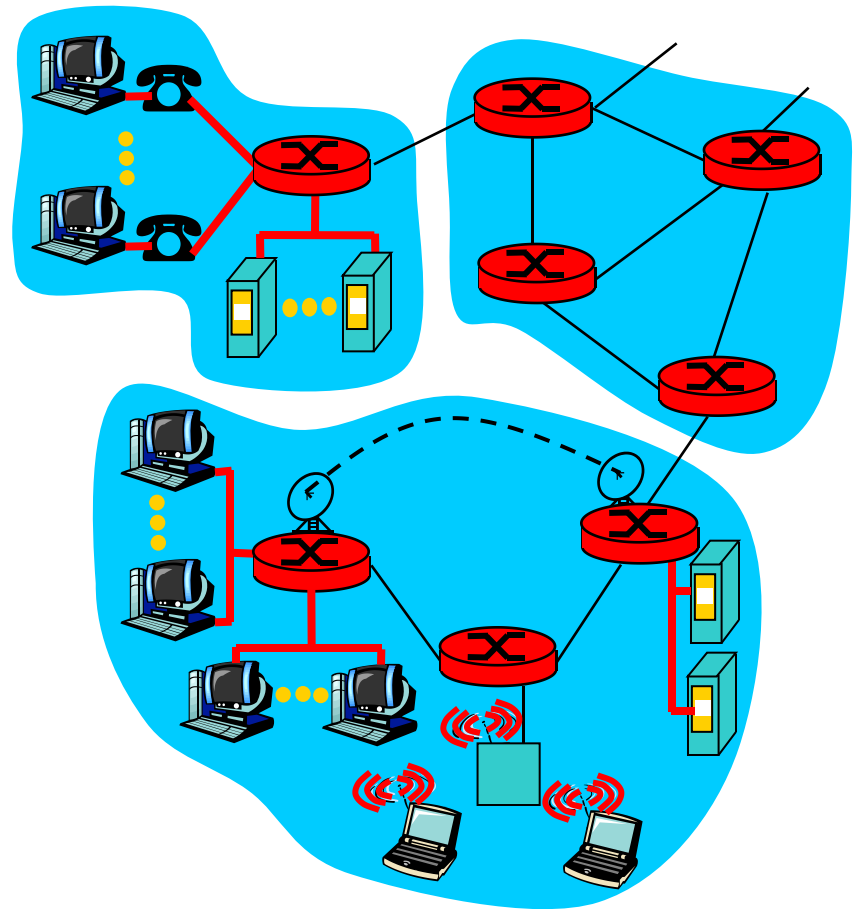
Access networks and physical media

Q: How to connect end systems to edge router?

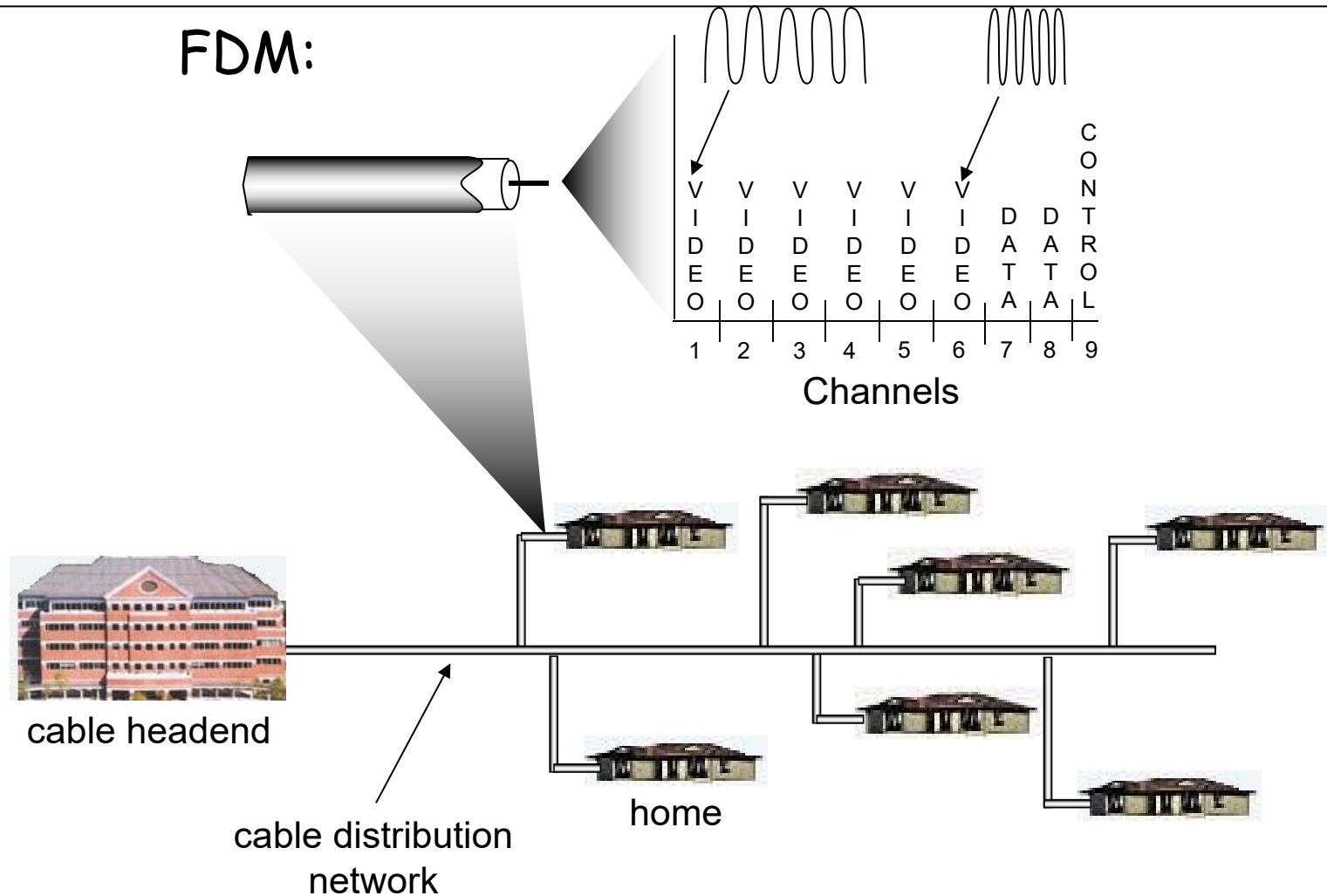
- residential access nets
- institutional access networks (school, company)
- mobile access networks

Keep in mind:

- bandwidth (bits per second) of access network?
- shared or dedicated?

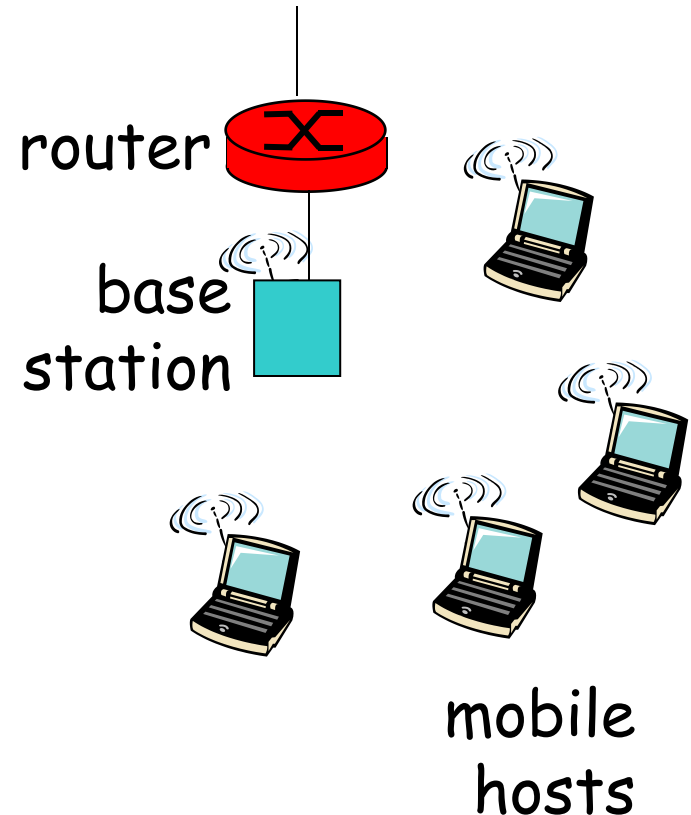


Cable Network Architecture:



Wireless access networks

- shared *wireless* access network connects end system to router
 - via base station AKA "access point"(AP hub)
- **wireless LANs:**
 - 802.11b/g (WiFi): 11 or 54 Mbps
- **wider-area wireless access**
 - provided by telco operator
 - 3G : 384 kbps
 - Will it happen??
 - GPRS in China, Europe/US



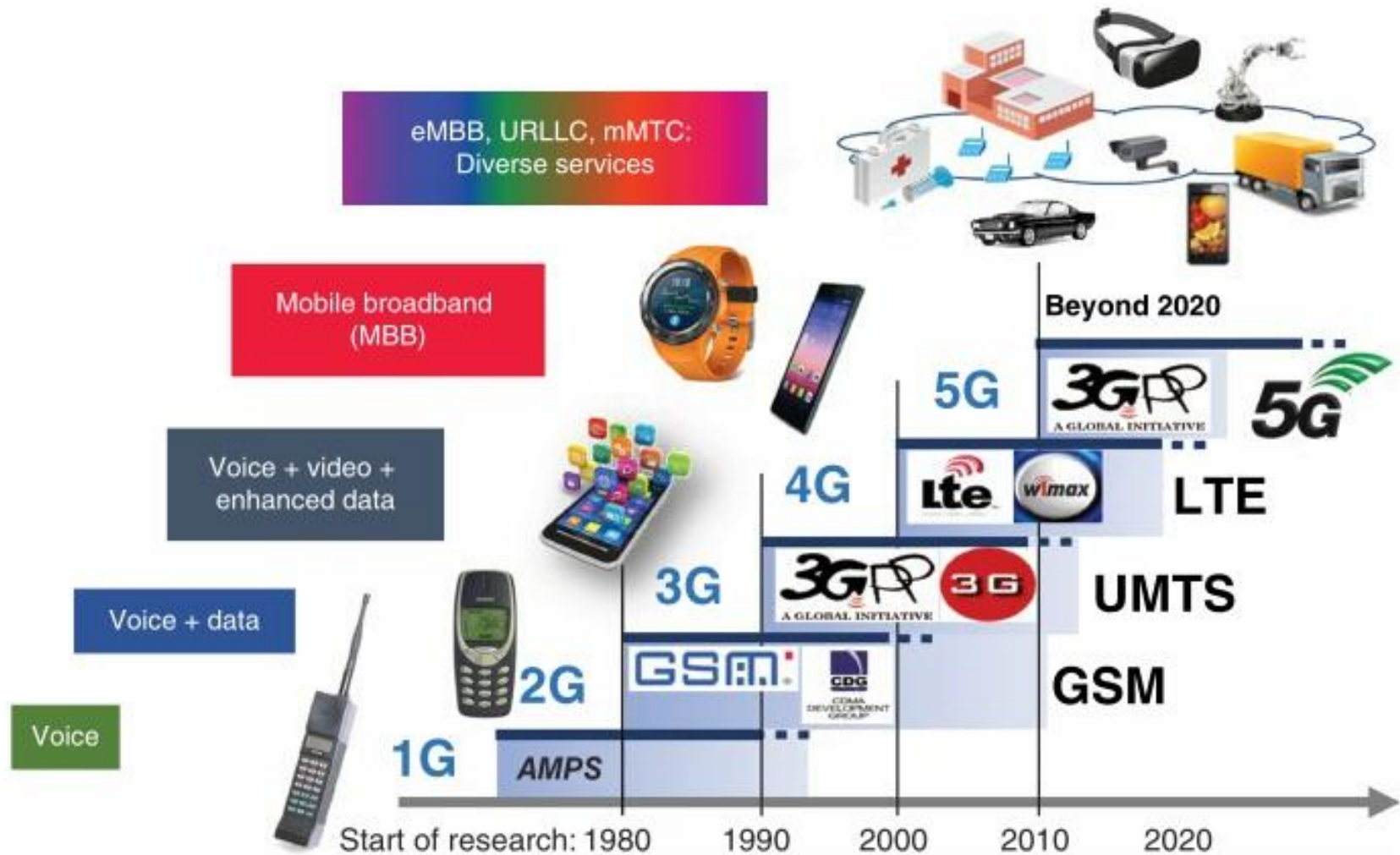
Physical media: radio

- signal carried in electromagnetic spectrum
- no physical “wire”
- bidirectional
- propagation environment effects:
 - reflection
 - obstruction by objects
 - interference

Radio link types:

- **Terrestrial microwave**
 - e.g. up to 45 Mbps channels
- **LAN** (e.g., Wifi)
 - 11Mbps, 54 Mbps, 1 Gbps
- **wide-area** (e.g., cellular)
 - e.g. 5G: 1 Gbps
- **satellite**
 - Kbps to 45Mbps channel (or multiple smaller channels)
 - 270 msec end-end delay

Mobile Networks History



1-3 PROTOCOLS AND STANDARDS

*In this section, we define two widely used terms: **protocols** and **standards**. First, we define protocol, which is synonymous with rule. Then we discuss standards, which are agreed-upon rules.*

Topics discussed in this section:

Protocols

Standards

Standards Organizations

Internet Standards

What's the Internet: a protocol view

What's a protocol?

human protocols:

- “what’s the time?”
- “I have a question”
- Introductions

... specific msgs sent

... specific actions taken when
msgs received, or other
events

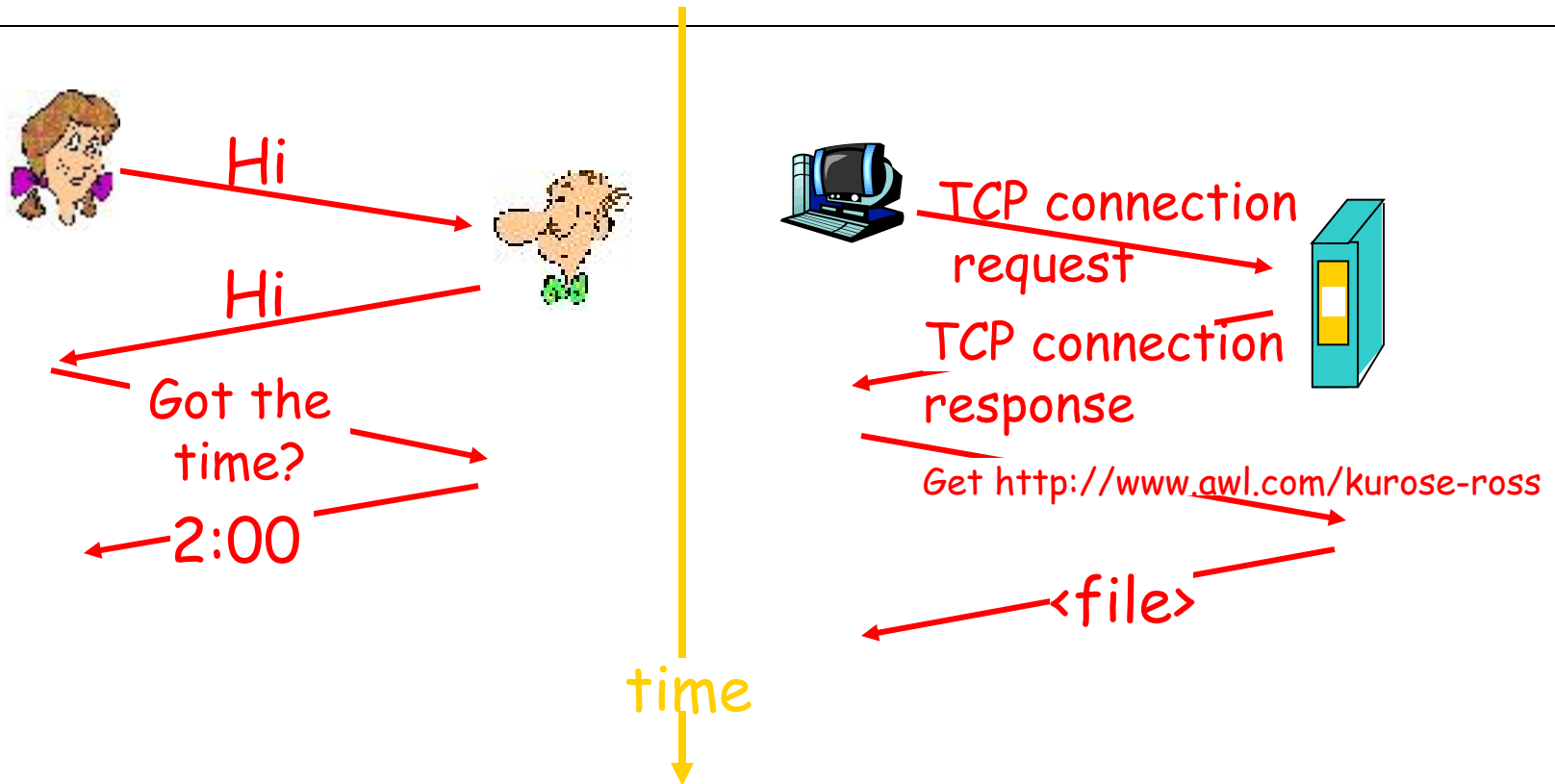
network protocols:

- machines rather than humans
- all communication activity in Internet governed by protocols

protocols define format, order of msgs sent and received among network entities, and actions taken on msg transmission, receipt

What's a protocol?

a human protocol and a computer network protocol:

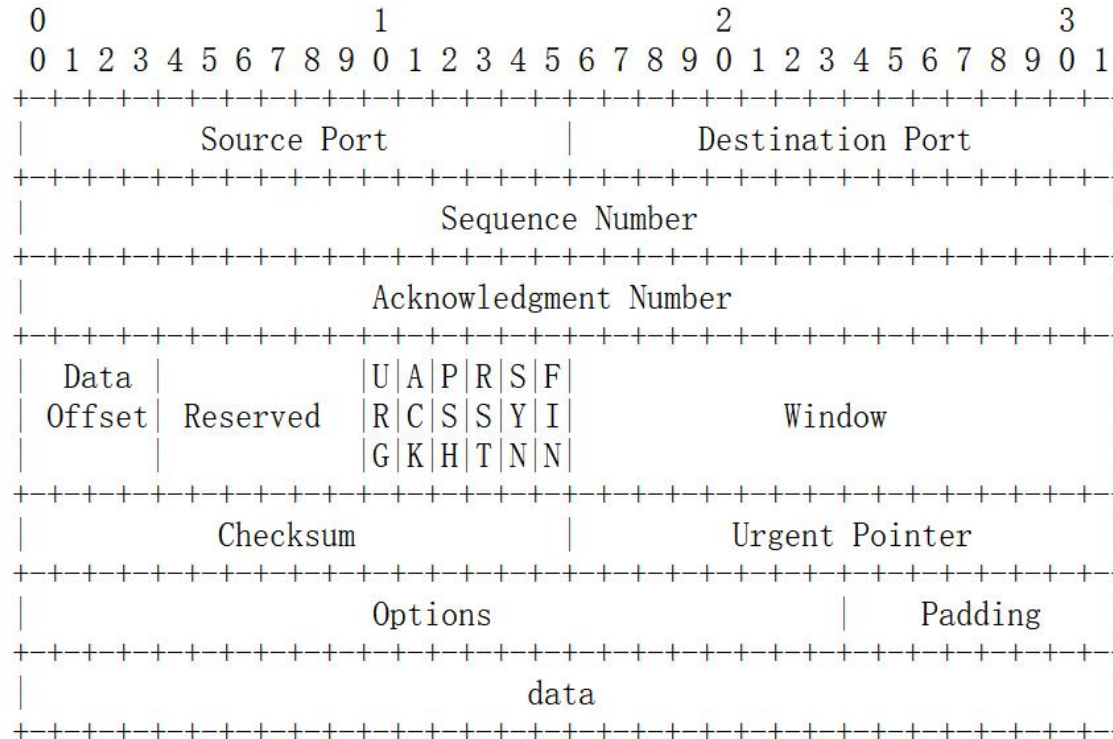


Q: Other human protocols?

A Protocol: Key Elements

- **Syntax:** refers to the structure or format of the data, meaning the order in which they are presented;
- **Semantics:** refers to the meaning of each section of bits;
- **Timing:** refers to two characteristics: when data should be sent and how fast they can be sent.

RFC 793 TCP Header Format

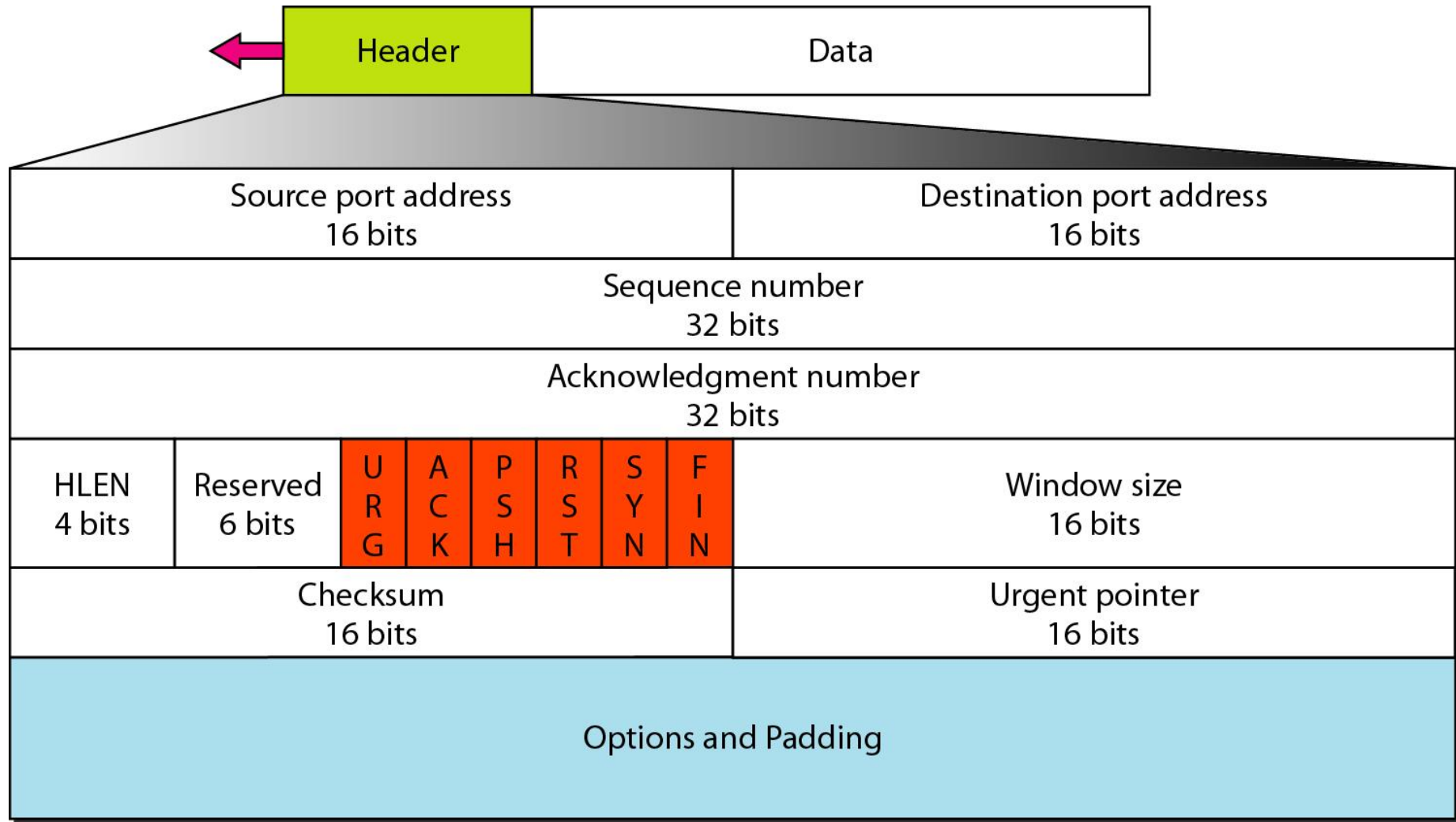


TCP Header Format

Note that one tick mark represents one bit position.

Figure 3.

Example *TCP segment format*



Standards: two categories

- De facto ("by fact" or "by convention")

Standards that have not been approved by an organization body but have been adopted as standards through widespread use

- De jure ("by law" or "by regulation")

Standards that have been legislated by an officially recognized body

Standards Organizations

- International Organization for Standardization (ISO)
- International Telecommunication Union-Telecommunication Standard Sector (ITU-T)
- 3rd Generation Partnership Project (3GPP)
- Institute of Electrical and Electronics Engineer (IEEE)
- American National Standards Institute (ANSI)
- Electronic Industries Association (EIA)
- ...

Internet Standards

- **An internet standard** is a thoroughly tested specification that is useful to and adhered to by those who work with the internet
- **An internet draft** is a working document with no official status and a 6-month lifetime.
- A draft may be published as a **Request for Comment (RFC)**.

References

- Behrouz A. Forouzan, Data Communications and Networking, 机械工业出版社, 2009.
- A. Osseiran, J.F. Monserrat, P. Marsch, 5G Mobile and Wireless Communications Technology, 中国工信出版集团, 2017.

Wireshark使用教程, 知乎/CSDN

- <https://zhuanlan.zhihu.com/p/654063052>
- <https://blog.csdn.net/dzqxwzoe/article/details/131763644>